

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Second Semester B.E. (Credit System) Degree Examinations
September - October 2022

21CY110 – ENGINEERING CHEMISTRY

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Derive the Nernst equation for the electrode reaction. $\text{Cu}^{2+} + 2\text{e}^- \rightleftharpoons \text{Cu}, \text{ at } 298^\circ\text{K}$	6	L*2	1	1
b)	An electrochemical cell consists of iron electrode dipped in 0.01 M FeSO ₄ solution and copper electrode dipped in 0.1 M CuSO ₄ solution. Write the cell representation, cell reaction and calculate the emf of the cell at 298 K. Given that standard reduction potential of iron and copper electrodes are -0.44 V and 0.34 V respectively.	6	L3	1	2
c)	What is ion-selective electrode? Explain the experimental method of determination of pH of a solution using glass electrode. Mention any two advantages.	8	L2	1	1
2. a)	Explain the electrochemical theory of corrosion taking iron as an example.	7	L2	1	1
b)	Write a note on anodized coating and galvanization.	7	L2	1	1
c)	Explain the following factors affecting the rate of corrosion: i) Complexing agent ii) pH	6	L2	1	1
3. a)	Describe the following battery characteristics: i) Voltage ii) Capacity iii) cycle life	8	L2	1	1
b)	Explain the construction, working of Ni-MH battery.	6	L2	1	1
c)	Define polarization. Describe any five factors affecting polarization.	6	L2	1	1
Unit – II					
4. a)	Describe the polymerization of vinyl chloride by free radical mechanism.	7	L2	2	1
b)	Explain the synthesis, application of following polymers: i) Polycarbonate ii) Epoxy resin	7	L2	2	1
c)	What are conducting polymers? Discuss the mechanism of conduction in poly-acetylene.	6	L2	2	1
5. a)	Explain the determination of dissolved oxygen by Winkler's method.	7	L2	2	1
b)	Explain Boiler Corrosion with reactions.	6	L2	2	1
c)	Describe the purification of water by reverse osmosis and electrodialysis.	7	L2	2	1

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|-------|--|---|----|---|---|
| 6. a) | What is glass transition temperature? Explain any five factors affecting the glass transition temperature. | 6 | L2 | 2 | 1 |
| b) | Describe the hot-lime soda process for softening of hard water. | 7 | L2 | 2 | 1 |
| c) | Explain the preparation of nano - particles by microwave and sol-gel method. | 7 | L2 | 2 | 1 |

Unit – III

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|-------|--|---|----|---|---|
| 7. a) | Explain the determination of calorific value of solid fuel using Bomb calorimeter with a neat diagram. | 7 | L2 | 3 | 1 |
| b) | What is petrol knocking in IC engine? Explain the mechanism of petrol knocking with reactions. | 7 | L2 | 3 | 1 |
| c) | Explain the electro-optic effect of liquid crystals. | 6 | L2 | 3 | 1 |
| 8. a) | A 0.6 g coal sample with 92% C, 5% H ₂ and 3% ash, caused a rise in the temperature of 2000 g of water by 3.2° C in a bomb calorimeter experiment. Calculate the gross and net calorific value of coal, given water equivalent = 200 g; Specific heat of water = 4.187 kJ/kg/°C; Latent heat of steam = 580 calories/g. | 6 | L3 | 3 | 2 |
| b) | What is octane number? Explain any five reactions involved in reformation. | 6 | L2 | 3 | 1 |
| c) | Describe with suitable examples, the liquid crystalline behavior of compounds based on their chemical constitutions. | 8 | L2 | 3 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
